

Lab 1 - Windows Event Log Investigation

Authentication Event Analysis in Windows Security Logs

Attribute	Details
Lab type	SOC / Windows Event Log Investigation
Environment	Windows 11 VM, VMware Workstation, Event Viewer
Key events	4625 - failed logon; 4624 - successful logon
Exercise type	Authorized controlled authentication test

1. Objective

Investigate Windows authentication activity, identify failed and successful logon events, and interpret the evidence using a basic SOC analyst workflow.

2. Procedure

- Locked the Windows 11 workstation and generated several controlled failed password attempts.
- Logged in successfully after the failed attempts.
- Opened Windows Event Viewer and reviewed the Security log.
- Filtered for Event IDs 4625 and 4624 and compared the event details.

3. Failed Logon - Event ID 4625

Field	Observation
Event ID	4625
Meaning	An account failed to log on
Activity	Controlled local failed sign-in
Analyst focus	Time, logon type, account context, source information, failure reason

The failed logon events were intentionally generated in the lab. A failed logon alone does not prove brute force or compromise; the analyst must review source, logon type, frequency, and surrounding activity.

4. Successful Logon - Event ID 4624

Field	Observation
Event ID	4624
Meaning	An account was successfully logged on
Use in investigation	Compare with prior failures and confirm whether authentication later succeeded

5. Findings

- Windows Security logging recorded both failed and successful authentication activity.
- Event IDs can be filtered to reconstruct a simple authentication sequence.
- Context is required before labeling repeated failures as malicious.
- The activity in this lab was controlled and did not indicate a real compromise.

6. Skills Demonstrated

- Windows Event Viewer
- Security Event ID analysis
- Authentication monitoring
- Basic SOC triage
- Evidence-based interpretation

7. Conclusion

The lab demonstrated practical investigation of Windows authentication telemetry. Event IDs 4625 and 4624 were used to distinguish failed and successful logons and to build a basic analyst interpretation without overstating the evidence.